Solutions to FIT3080 Applied Class on Propositional and First-order Logic

Exercise 1: Propositional logic – models, validity and satisfiability

Consider a vocabulary with only four propositions A, B, C and D. How many models are there in the following sentences? Are these sentences valid, satisfiable (but not valid) or unsatisfiable?

(a) $B \lor C$

SOLUTION:

The sentence is false only if B and C are false, which occurs in 4 cases for all the combinations of A, B, C and D, leaving 12 models.

Another way to look at this is:
A={T,F}, B=T, C=F, D={T,F}: 4 models
A={T,F}, B=F, C=T, D={T,F}: 4 models
A={T,F}, B=T, C=T, D={T,F}: 4 models
Total: 12 models.

It doesn't hold in four models: A={T,F}, B=F, C=F, D={T,F}

So, this sentence is satisfiable but not valid.

(b) $\neg A \lor \neg B \lor \neg C \lor \neg D$

SOLUTION:

The sentence is false only if A, B, C, and D are true, which occurs in 1 case, leaving 15 models.

So, this sentence is satisfiable but not valid (doesn't hold in one model).

Exercise 2: Propositional logic – validity and satisfiability

Consider the following sentence:

$[(Food \Rightarrow Party) \lor (Drink \Rightarrow Party)] \Rightarrow [(Food \land Drink) \Rightarrow Party]$

Convert this sentence to CNF, and determine whether it is valid, satisfiable (but not valid) or unsatisfiable.

SOLUTION:

$[(Food \Rightarrow Party) \lor (Drink \Rightarrow Party)] \Rightarrow [(Food \land Drink) \Rightarrow Party]$
$\neg[(\neg Food \lor Party) \lor (\neg Drink \lor Party)] \lor [\neg(Food \land Drink) \lor Party]$
$\neg[\neg Food \lor \neg Drink \lor Party] \lor [\neg Food \lor \neg Drink \lor Party] \equiv True$
So, the sentence is **valid**.

Exercise 3: Validity and satisfiability

Prove that $\alpha \models \beta$ iff $\alpha \land \neg\beta$ is unsatisfiable. This is the main theorem for resolution-refutation.

SOLUTION:

The Deduction Theorem (Slide 21) states that $\alpha \models \beta$ iff $\alpha \Rightarrow \beta$ is valid.

(Slide 22) A sentence is valid (true in all models) iff its negation is unsatisfiable (false in all models).

Say our sentence is $\alpha \Rightarrow \beta$, then $\alpha \Rightarrow \beta$ is valid iff $\neg(\alpha \Rightarrow \beta)$ is unsatisfiable

$\neg(\alpha \Rightarrow \beta) \equiv \neg(\neg\alpha \lor \beta) \equiv \alpha \land \neg\beta$, which proves our assertion.

Let's now prove the Deduction Theorem.

Start from $\alpha \models \beta$. This means that β is true in every model where α is true. We now have two cases:

- α is true: therefore β is true (by entailment), and $\alpha \Rightarrow \beta$ is true in all models where α is true.
- α is false: $\alpha \Rightarrow \beta$ is true in all models where α is false (because of the definition of implication).

So, the implication is true in all models, i.e., $\alpha \Rightarrow \beta$ is valid.

Start from $\alpha \Rightarrow \beta$ is valid. This means that the implication is true in all models. Again, let's consider two cases:

- α is true: then β must be true (because the implication is true). So β is true in every model where α is true, which means that $\alpha \models \beta$.
- α is false: not interesting for proving entailment.

Exercise 4: Propositional Logic – Horn clauses, forward and backward reasoning

Consider the following statements:

1. If Susan gets high marks she is successful.
2. If Susan is bright and studies she will get high marks.
3. If Susan is not bright she won't pass the subject.
4. If Susan is diligent she will study.
5. If Susan is not diligent she will have fun.
6. If Susan has fun she will not study.
7. Susan is diligent.
8. Susan has passed the subject.

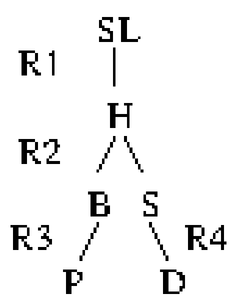
(a) Using the propositions H (high marks), SL (successful), B (bright), S (study), P (pass), D (diligent) and F (have fun), translate the above statements to Propositional Logic. Which of these clauses can be converted to Horn clauses?

SOLUTION:

R1. $H \Rightarrow SL$	$\neg H \vee SL$
R2. $(B \wedge S) \Rightarrow H$	$\neg B \vee \neg S \vee H$
R3. $\neg B \Rightarrow \neg P$	$B \vee \neg P$ – change to $P \Rightarrow B$
R4. $D \Rightarrow S$	$\neg D \vee S$
R5. $\neg D \Rightarrow F$	$D \vee F$ – NOT HORN CLAUSE
R6. $F \Rightarrow \neg S$	$\neg F \vee \neg S$
$*\ D$	D
$*\ P$	P

(b) Apply backward reasoning to prove that Susan is successful.

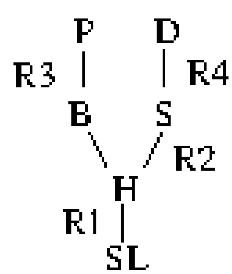
SOLUTION:



(c) Apply forward reasoning to prove that Susan is successful.

SOLUTION:

Queue	Count					Inferred
	R1	R2	R3	R4	R6	
D, P	1	2	1	1	1	
D, B	1	2	0	1	1	P
B, S	1	2	0	0	1	P, D
S	1	1	0	0	1	P, D, B
H	1	0	0	0	1	P, D, B, S
SL	0	0	0	0	1	P, D, B, S, H



Exercise 5: First-order Logic – Equivalence

Prove or argue that the following equivalences hold (Slide 48):

(a) $\forall x[P(x) \wedge Q(x)] \equiv \forall x P(x) \wedge \forall y Q(y)$

SOLUTION:

For all entities x, $P(x)$ is true and $Q(x)$ is true. So it doesn't matter how we run the counters. Specifically, we can run one loop and count all the xs and the ys, or we can run one loop that counts all the xs and another than counts all the ys.

(b) $\exists x[P(x) \vee Q(x)] \equiv \exists x P(x) \vee \exists y Q(y)$

SOLUTION:

For this statement to be true, there must exist an x such that $P(x)$ is true or $Q(x)$ is true.

Start with $\exists x[P(x) \vee Q(x)]$**.** Assume $P(x)$ is true, then $\exists x P(x)$. This sentence can be OR-ed with anything (e.g., $\exists y Q(y)$), which yields $\exists x P(x) \vee \exists y Q(y)$.

Start with $\exists x P(x) \vee \exists y Q(y)$**.** Assume $\exists x P(x)$ is true. The $P(x)$ can be OR-ed with anything, which yields $\exists x[P(x) \vee Q(x)]$.

The same can be done for Q.

Exercise 6: First-order Logic – Unification

In the following pairs of expressions, w, x, y and z are variables, A and B are constants, and f and g are functions. Do these pairs of expressions unify? If they do, give the substitution that unifies them. If they don't, then indicate where the unification fails.

(a) $P(x, f(x))$ and $P(f(A), f(A))$

SOLUTION:

To unify the first term in both expressions, we need to substitute $\{x|f(A)\}$. This results in $P(f(A), f(f(A)))$ and $P(f(A), f(A))$. Now, the second term cannot unify, because $f(A)$ does not unify with A.

UNIFICATION FAILS.

(b) $P(x, f(x), f(x))$ and $P(y, z, f(y))$

SOLUTION:

Substituting $\{y|x\}$ we get $P(x, f(x), f(x))$ and $P(x, z, f(x))$. Now,

Substituting $\{z|f(x)\}$ we get $P(x, f(x), f(x))$ and $P(x, f(x), f(x))$. The mgu is $\{y|x, z|f(x)\}$.

Exercise 7: First-order Logic – Resolution refutation

Consider the following statements:

1. John likes all food.
2. Anything that one eats and isn't killed by is food.
3. Bill eats peanuts.
4. Bill is still alive.

(a) Represent these statements as predicate calculus formulas using the predicates: LIKES, FOOD, EATS and ALIVE.

SOLUTION:

1. $\forall x\ [\ FOOD(x) \Rightarrow LIKES\ (John,x)\ ]$
2. $\forall x \forall y\ [\ EATS(y,x) \wedge ALIVE(y)\ \Rightarrow FOOD(x)\ ]$
3. $EATS(Bill,peanuts)$
4. $ALIVE(Bill)$

(b) Convert these statements to clauses.

SOLUTION:

1. $\forall x\ [\ FOOD(x) \Rightarrow LIKES\ (John,x)\ ]$

- 1. Eliminate $\Rightarrow$: $\forall x\ [\ \neg FOOD(x) \vee LIKES\ (John,x)\ ]$
- 7. Eliminate $\forall$: $\neg FOOD(x) \vee LIKES\ (John,x)$

2. $\forall x \forall y\ [\ EATS(y,x) \wedge ALIVE(y)\ \Rightarrow FOOD(x)\ ]$

- 1. Eliminate $\Rightarrow$: $\forall x \forall y\ [\ \neg\ (\ EATS(y,x) \wedge ALIVE(y)\ )\ \vee FOOD(x)\ ]$
- 2. Reduce scopes of $\neg$: $\forall x \forall y\ \neg EATS(y,x) \vee \neg ALIVE(y) \vee FOOD(x)$
- 7. Eliminate $\forall$: $\neg EATS(y,x) \vee \neg ALIVE(y) \vee FOOD(x)$

3. $EATS(Bill,peanuts)$
4. $ALIVE(Bill)$

The final step consists of standardizing variables apart for each clause (Step 9):

1. $\neg FOOD(x1) \vee LIKES\ (John,x1)$
2. $\neg EATS(y,x2) \vee \neg ALIVE(y) \vee FOOD(x2)$

No changes are needed for clauses 3 and 4.

(c) Use resolution-refutation to prove that John likes peanuts.

SOLUTION:

Goal: $LIKES(John,peanuts)$

Negated goal: 5. $\neg LIKES(John,peanuts)$

5 and 1: 5. $\underline{\neg LIKES(John,peanuts)}$ 1. $\neg FOOD(x1) \vee \underline{LIKES(John,x1)}$

mgu: $\{x1|\text{peanuts}\}$

resolvent: 6. $\neg$FOOD(peanuts)

6. $\underline{\neg\text{FOOD(peanuts)}}$ 2. $\neg EATS(y,x2) \vee \neg ALIVE(y) \vee \underline{FOOD(x2)}$

mgu: $\{x2|\text{peanuts}\}$

resolvent: 7. $\neg$EATS(y,peanuts) $\vee$ $\neg$ALIVE(y)

7. $\underline{\neg\text{EATS}(y,\text{peanuts})} \vee \neg\text{ALIVE}(y)$ 3. $EATS(Bill,peanuts)$

mgu: $\{y|\text{Bill}\}$

resolvent: 8. $\neg$ALIVE(Bill)

8. $\neg$ALIVE(Bill) 4. $ALIVE(Bill)$

resolvent: NIL